

COAQNN-Based Real-Time NIFTY-50 Stock Price Prediction Using Financial Indicators and Sentiment Analysis

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Abstract— Stock market prediction is a complex task due to the highly volatile and non-linear nature of financial markets. Accurate forecasting of stock prices can help investors make informed financial decisions and reduce investment risks. This research proposes a COAQNN-based stock price prediction system for NIFTY-50 stocks using real-time financial data and deep learning techniques. Historical and live stock market data are collected using the Yahoo Finance API and processed using technical indicators such as RSI, EMA, MACD, volatility, moving averages, and lag features. The proposed COAQNN (Coati Optimization Algorithm–Quantum Neural Network) model predicts both mid and closing stock prices with improved accuracy. The system performance is evaluated using MAE, RMSE, MAPE, and Accuracy, and compared with Random Forest Regression and Linear Regression models. Experimental results demonstrate that the proposed COAQNN model achieved an accuracy of 98.27%, outperforming several existing approaches from the literature survey, including Logistic Regression-based prediction models with 68.62% accuracy and other hybrid forecasting methods. A Streamlit-based web application is also developed for real-time stock price prediction and analysis.

Keywords— Stock Price Prediction, COAQNN, Deep Learning, NIFTY-50, Financial Forecasting, Technical Indicators, Machine Learning, Streamlit, Yahoo Finance API

I. Introduction

The stock market is one of the most important components of the global financial system and plays a significant role in investment, portfolio management, and economic growth. Predicting stock prices accurately is a difficult task because stock market behavior is highly dynamic, volatile, and influenced by several financial, political, and economic factors. Traditional statistical methods often fail to capture the complex and non-linear relationships present in financial time-series data, resulting in lower prediction accuracy.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have significantly improved stock market forecasting techniques. Deep learning models can learn hidden patterns from large amounts of historical and real-time financial data, enabling more accurate stock price prediction. Technical indicators such as Relative Strength Index (RSI), Exponential Moving Average (EMA), Moving Average Convergence Divergence (MACD), volatility, moving averages, and lag features further enhance prediction performance by extracting meaningful market trends and behavioral patterns.

This research proposes a COAQNN-based real-time stock price prediction system for NIFTY-50 stocks using financial indicators and Sentiment Analysis. Historical and live stock market data are collected using the Yahoo Finance API and processed through data preprocessing and feature engineering techniques. The proposed COAQNN model predicts both mid and closing stock prices and is compared with Random Forest Regression and Linear Regression models. Experimental results show that the proposed model achieved an accuracy of 98.27%, outperforming several traditional machine learning approaches. A Streamlit-based web application is also developed to provide an interactive platform for real-time stock price prediction and analysis. The main contributions of this research can be summarized as follows:

1. Development of a hybrid COAQNN-based deep learning framework for real-time NIFTY-50 stock price prediction using financial indicators and historical market data.
2. Integration of technical indicators such as RSI, EMA, MACD, volatility, moving averages, and lag features to improve feature representation and prediction accuracy.

3. Implementation of a dual prediction mechanism capable of forecasting both mid stock prices and closing stock prices using real-time financial data collected from the Yahoo Finance API.
4. Comparative performance analysis of the proposed COAQNN model with traditional machine learning models such as Random Forest Regression and Linear Regression using MAE, RMSE, MAPE, and Accuracy.
5. Development of a Streamlit-based real-time stock prediction web application that enables interactive forecasting and visualization for investors and financial analysts.

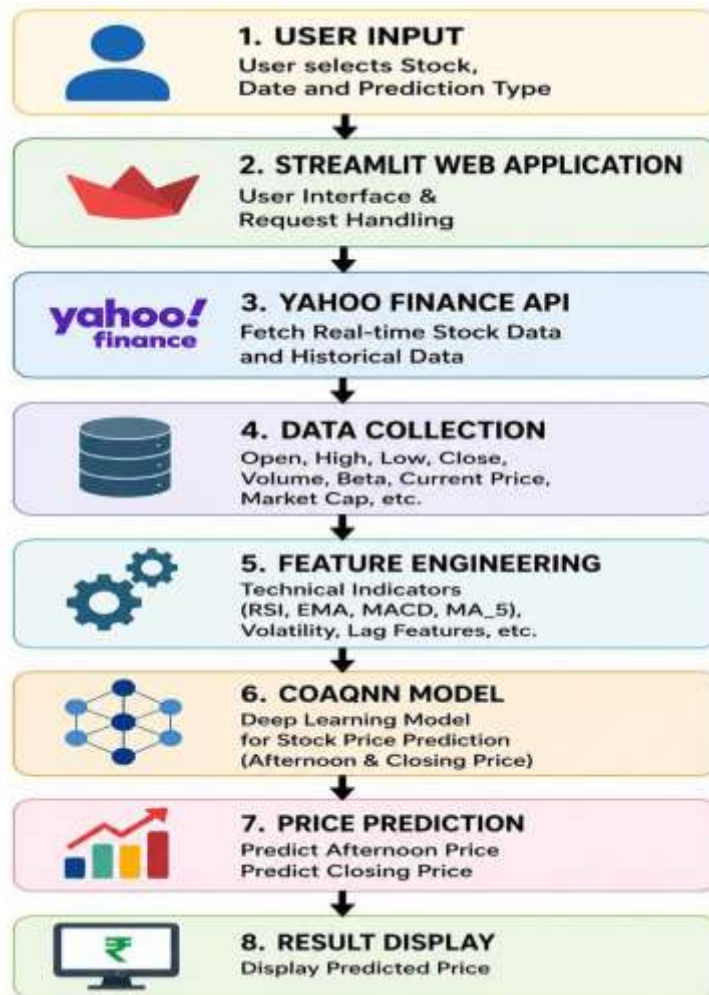


Fig 1. The general process of Stock Price Prediction

II. Literature Review

Traditional stock market prediction methods mainly relied on statistical analysis and mathematical forecasting techniques. However, due to the highly dynamic, volatile, and non-linear behavior of financial markets, traditional approaches often produced lower prediction accuracy and failed to capture complex market patterns. In recent years, significant advancements have been made in the field of stock market forecasting using Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) techniques. Researchers have developed various intelligent models capable of learning hidden temporal dependencies and improving prediction performance using historical financial data and technical indicators. An extensive literature survey was conducted on stock price prediction methods proposed using machine learning, deep learning, and hybrid forecasting techniques. Some of the important research works are discussed below.

Shuzhen Wang [1] proposed a hybrid stock prediction framework combining Bidirectional Long Short-Term Memory (BiLSTM), an improved Transformer architecture, and Temporal Convolutional Networks (TCN). The proposed framework effectively captured both short-term and long-term temporal dependencies present in financial time-series data. The improved Transformer architecture enhanced feature extraction and sequence learning capability, while the TCN component improved model

generalization and forecasting stability. Experimental results demonstrated significant improvements in prediction accuracy, RMSE, and R^2 score compared to existing methods. However, the proposed model required high computational resources and increased training complexity due to the integration of multiple deep learning architectures.

Srinath Ravikumar and Prasad Saraf [2] developed stock prediction models using regression and classification algorithms for financial forecasting. Historical stock market data obtained from Yahoo Finance was preprocessed using feature engineering techniques such as momentum and volatility extraction. The study utilized regression models to predict future stock prices and classification algorithms to identify market movement direction. Logistic Regression achieved a prediction accuracy of 68.62%. The research highlighted the importance of technical indicators and preprocessing techniques in stock prediction systems. However, the model showed limited performance under highly volatile market conditions.

Eti Khatri et al. [3] introduced a hybrid deep learning framework combining Long Short-Term Memory (LSTM) and Bidirectional Gated Recurrent Unit (BiGRU) networks for stock price prediction and financial trend analysis. The proposed model effectively captured sequential dependencies and improved the analysis of risk-return relationships in stock market data. The framework demonstrated improved forecasting performance compared to traditional machine learning models and achieved better prediction accuracy for technology sector stocks. Despite the improved performance, the model introduced higher computational overhead and increased training time due to the use of multiple recurrent neural network architectures.

Mohammad Raquibul Hossain, Mohd Tahir Ismail, and Samsul Ariffin Bin Abdul Karim [4] proposed an EMD-ARIMA-EWMA hybrid forecasting model for stock market prediction. The study utilized Empirical Mode Decomposition (EMD) to separate high-frequency and low-frequency stock market components before applying ARIMA and EWMA forecasting methods. The proposed hybrid framework improved prediction accuracy and outperformed traditional statistical forecasting approaches across multiple datasets and forecasting horizons. However, the model struggled to effectively handle highly non-linear financial market behavior and required careful parameter tuning for different datasets.

The literature survey indicates that deep learning and hybrid forecasting models significantly improve stock market prediction accuracy compared to traditional statistical approaches. However, several existing systems suffer from limitations such as high computational complexity, limited real-time adaptability, insufficient integration of technical indicators, and lower prediction accuracy in highly volatile market environments. To address these limitations, this research proposes a COAQNN-based real-time stock price prediction system integrated with technical indicators, real-time financial APIs, and deep learning techniques for accurate NIFTY-50 stock forecasting.

III. Methodology

The methodology describes the systematic approach adopted for stock price prediction using the proposed COAQNN (Coati Optimization Algorithm - Quantum Neural Network) model. This section covers data collection and preprocessing, feature engineering, dataset preparation, COAQNN architecture design, optimization, model training, and prediction. The objective is to leverage historical stock market data, technical indicators, analyst recommendations, and news sentiment information to accurately forecast future stock prices.

3.1 Data Collection and Preprocessing: Historical and real-time stock market data for NIFTY-50 companies were collected using the Yahoo Finance API. The dataset includes important financial attributes such as Open, High, Low, Close, Volume, Beta, Current Price, and Recommendation values. Two separate datasets were prepared for the prediction tasks. The mid stock price prediction dataset contained 3790 records, while the closing stock price prediction dataset contained 2490 records.

During preprocessing, missing and null values were removed to improve dataset quality and consistency. Feature scaling and normalization were performed using the MinMaxScaler technique to transform all feature values into a common numerical range, thereby improving model convergence and prediction stability. The normalization process is represented as:

$$X_{\text{scaled}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

Where X denotes the original feature value, X_{min} and X_{max} represent the minimum and maximum values of the respective feature in the dataset, and X_{scaled} is the normalized value obtained after scaling. This transformation ensures that features with larger numerical ranges do not dominate the learning process and allows the COAQNN model to train more effectively.

This preprocessing stage improved the learning efficiency of the COAQNN model and reduced the impact of data imbalance and feature magnitude variations during training.

Feature Engineering: Feature engineering was performed to improve the prediction capability and learning efficiency of the proposed COAQNN model. Various technical indicators and derived financial attributes were generated from the stock market dataset to capture hidden market trends, momentum, volatility, and temporal dependencies. Important indicators such as Relative Strength Index (RSI), Exponential Moving Average (EMA), Moving Average Convergence Divergence (MACD), volatility, price range, moving averages, and lag features were extracted from the historical stock data. The RSI indicator was used to measure market momentum and identify overbought or oversold conditions, while EMA and moving averages helped in identifying stock price trends over time. MACD was utilized to analyze trend direction and momentum changes in the market. Volatility and price range features captured daily market fluctuations, and lag features were introduced to learn temporal dependencies from previous stock prices. These engineered features significantly improved the forecasting accuracy and robustness of the proposed stock price prediction system.

Data Splitting: The dataset was divided into separate subsets for training and testing the proposed COAQNN model. The dataset was partitioned carefully to ensure diversity and proper representation of stock market patterns in each subset. Approximately 80% of the dataset was used for training, while the remaining 20% was reserved for testing and evaluation purposes. The training dataset was utilized to train the COAQNN model and learn complex financial patterns from historical stock data. The testing dataset was used to evaluate the prediction performance and generalization capability of the model on unseen data. This structured data splitting process helped in reducing overfitting and ensured reliable and accurate stock price forecasting results.

3.2 COAQNN Architecture Design: The proposed COAQNN (Coati Optimization Algorithm–Quantum Neural Network) architecture plays a major role in accurately predicting mid and closing stock prices using historical and real-time financial data. The model is designed to capture complex non-linear financial patterns, temporal dependencies, and market volatility through deep learning and optimization techniques. The overall COAQNN architecture consists of several important layers and components that work together to improve prediction accuracy and forecasting stability.

Input Layer: The Input Layer serves as the entry point of the proposed model. This layer accepts the preprocessed financial dataset containing engineered features such as Open, High, Low, Close, Volume, RSI, EMA, MACD, volatility, moving averages, lag features, Beta, and recommendation values. All input features are normalized using MinMaxScaler before being passed into the neural network to improve learning efficiency and convergence performance.

COA Hyperparameter Optimization Layer: The Coati Optimization Algorithm (COA) is used to determine the optimal hyperparameters for the Quantum Neural Network. It searches for the best configuration of parameters such as the number of neurons in hidden layers, learning rate, and weight initialization values. This optimization process improves model convergence, prediction stability, and forecasting accuracy.

Quantum Neural Network (QNN) Layer: The Quantum Neural Network forms the core prediction engine of the proposed system. It consists of multiple dense hidden layers that learn complex and nonlinear relationships between financial indicators and stock price movements. The tanh activation function is applied within the hidden layers to introduce non-linearity and improve pattern recognition capabilities. The QNN effectively captures market trends, volatility patterns, and historical dependencies present in stock market data.

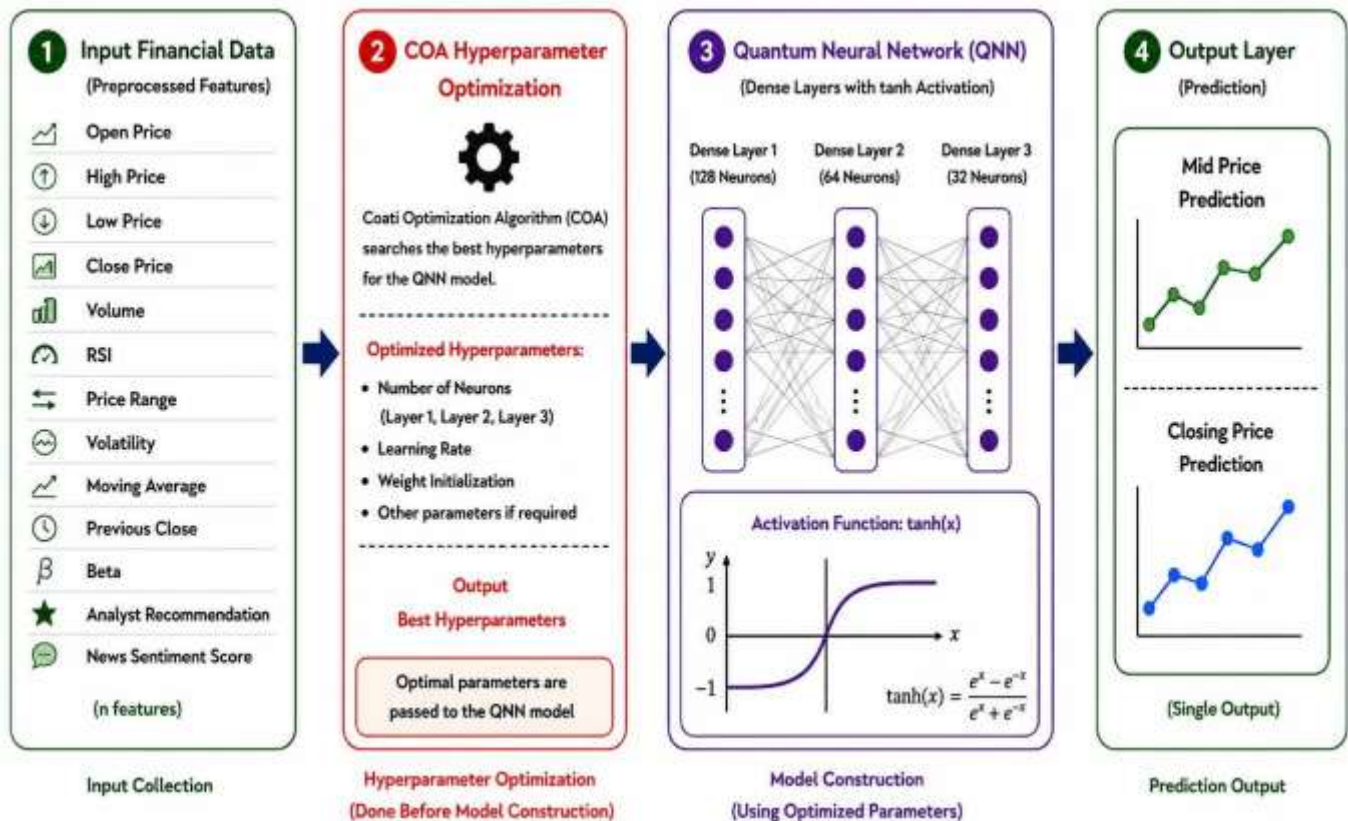


Fig 2. Fundamental Architecture of COAQNN

Output Layer: The Output Layer generates the final stock price forecasts based on the learned patterns from the QNN. The model produces predictions for both Mid Price and Closing Price, enabling investors and users to analyze future market behavior and make informed financial decisions.

3.3 Model Training: The proposed COAQNN model was trained using the prepared NIFTY-50 stock market dataset containing historical and real-time financial features. During the training process, the model learned complex relationships between technical indicators and stock price movements for both mid-price and closing-price prediction tasks. The Mean Squared Error (MSE) loss function was used to measure the difference between predicted and actual stock prices. To minimize the prediction error and improve convergence efficiency, the Adam optimization algorithm was utilized. The model was trained using multiple epochs and batch processing techniques to improve learning stability, forecasting accuracy, and generalization capability under volatile market conditions.

3.4 Implementation

The implementation of the proposed COAQNN-based stock price prediction system is an important stage that transforms the theoretical model into a practical real-time forecasting application. This phase includes dataset preparation, feature engineering, model development, training, evaluation, and deployment through a user-friendly web interface.

Data Collection: Historical and real-time stock market data of NIFTY-50 companies were collected using the Yahoo Finance API. The dataset included financial attributes such as Open, High, Low, Close, Volume, Beta, Current Price, and Recommendation values for both mid-price and closing-price prediction tasks.

Data Preprocessing: The collected dataset was preprocessed by removing missing and inconsistent values. Feature normalization was performed using the MinMaxScaler technique to improve model convergence and prediction stability.

Split Dataset: The prepared dataset was divided into training and testing subsets using an 80:20 ratio. The training dataset was used for model learning, while the testing dataset was used for performance evaluation and validation.

Building the COAQNN Model: The architecture of the COAQNN model was designed specifically for stock market forecasting. The model consisted of input layers, dense fully connected layers, activation layers, optimization components, and output prediction layers. The dense layers learned complex financial patterns from engineered features, while the tanh activation function introduced non-linearity for improved learning capability. The Coati Optimization Algorithm (COA) was integrated to optimize model parameters and improve prediction stability and convergence speed.

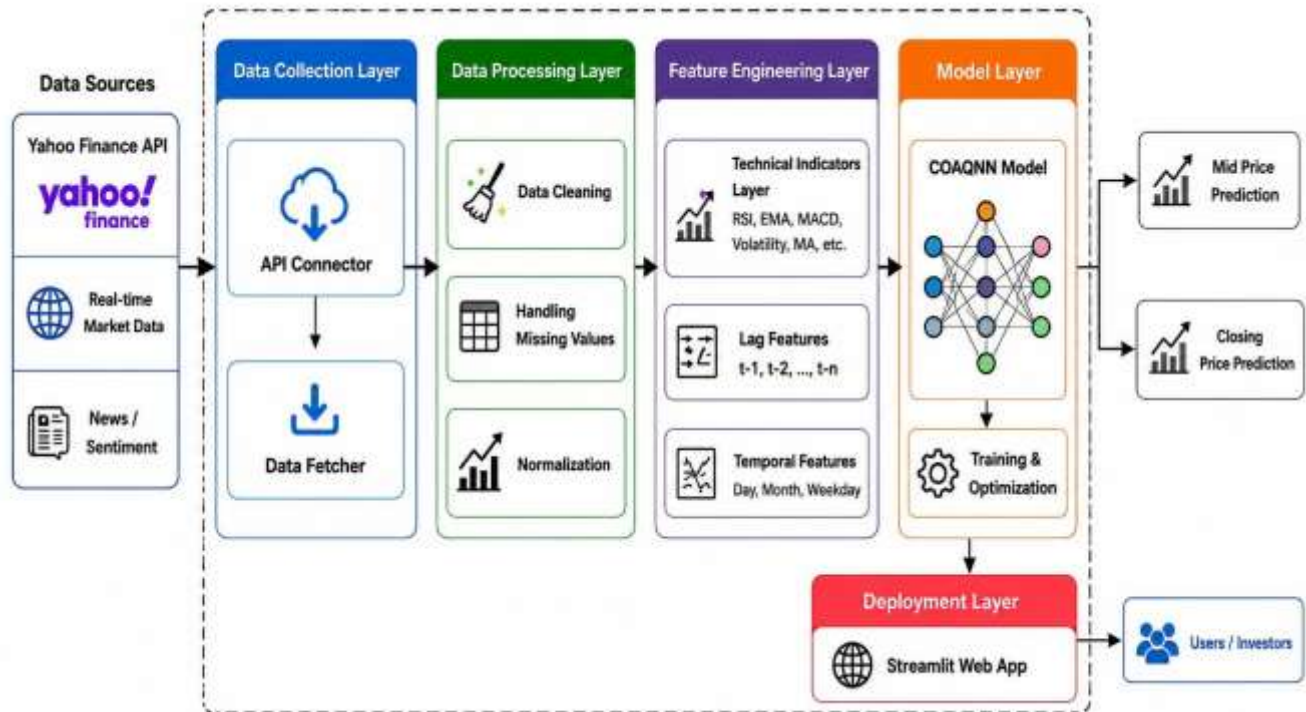


Fig 3. System Architecture.

COAQNN Model Training: The COAQNN model was trained using the prepared training dataset for both mid-price prediction and closing-price prediction tasks. The Adam optimizer and Mean Squared Error (MSE) loss function were used during training to minimize forecasting errors. The model was trained for multiple epochs using a batch size of 8 to achieve higher prediction accuracy and reduced RMSE values.

Validation and Model Evaluation: The trained COAQNN model was evaluated using various performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and Accuracy. The proposed model was also compared with traditional machine learning algorithms such as Random Forest Regression and Linear Regression to validate its performance improvement.

Model Testing: Evaluating the final trained model using the test dataset to ensure its performance on unseen data and validate its ability to predict stock price accurately.

User Interface Development: A user-friendly web application was developed using Streamlit for real-time stock market prediction. The application allows users to select a NIFTY-50 stock, choose the prediction date, and select prediction categories such as mid-price prediction, closing-price prediction. The system then generates real-time stock price forecasts using the trained COAQNN model.



Stock Price Prediction System

COAQNN Based Stock Forecasting

Select Stock

RELIANCE.NS

Select Date

2026/05/24

Select Prediction Type

Mid Price

Predict Price

Fig 4. User Interface of Stock Price Prediction Website

System Integration and Deployment: Finally, the trained COAQNN model, preprocessing modules, feature engineering components, and Streamlit web interface were integrated into a complete stock forecasting system. The deployed system provides an interactive and efficient platform for investors and financial analysts to perform real-time stock price prediction and analysis.

IV. Results

To evaluate the effectiveness of the proposed COAQNN-based stock price prediction system, a comprehensive performance analysis was conducted using real-time and historical NIFTY-50 stock market data. The evaluation focused on measuring the model's ability to accurately predict both mid stock prices and closing stock prices under dynamic market conditions. The prediction results generated by the COAQNN model were compared with actual stock market prices to assess forecasting accuracy and reliability.

The performance of the proposed model was analyzed using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Accuracy, and R^2 Score. Experimental results demonstrated that the COAQNN model achieved higher prediction accuracy and lower error rates compared to traditional machine learning models such as Random Forest Regression and Linear Regression. The model effectively captured hidden financial patterns, temporal dependencies, and market volatility through the integration of technical indicators such as RSI, EMA, MACD, moving averages, volatility, and lag features.

The figure 5 illustrates the comparison between the actual Mid stock prices and the prices predicted by the proposed COAQNN model across multiple test samples. The close alignment between the actual and predicted curves demonstrates the model's ability to accurately capture underlying market trends and price fluctuations. The predicted values closely follow the actual stock prices, including periods of significant rises and declines, indicating that the model effectively learns complex nonlinear relationships from technical indicators, sentiment scores, and historical market data.

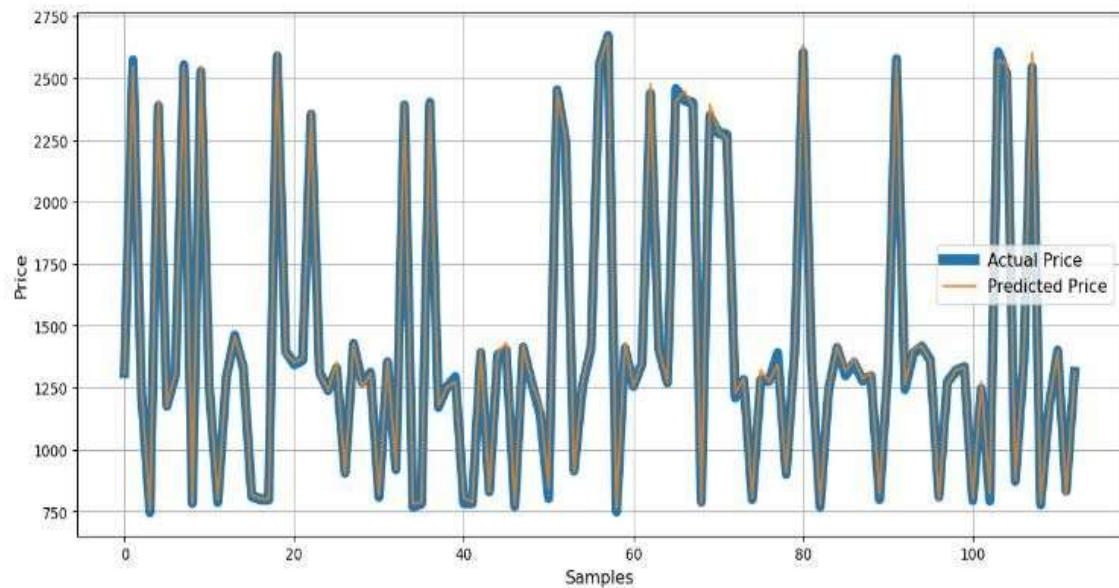


Fig 5. Actual vs Predicted – Mid Stock Price

The Figure 6 presents the scatter plot comparison between the actual Mid stock prices and the prices predicted by the proposed COAQNN model. Each point represents a test sample, where the x-axis denotes the actual stock price and the y-axis denotes the corresponding predicted price. The clustering of points along a near-linear diagonal trend indicates a strong positive correlation between actual and predicted values, demonstrating the model's high predictive accuracy. The close proximity of most data points suggests that the COAQNN model effectively captures underlying market patterns and minimizes prediction errors. This strong agreement between actual and predicted prices confirms the robustness and reliability of the proposed model for stock price forecasting.

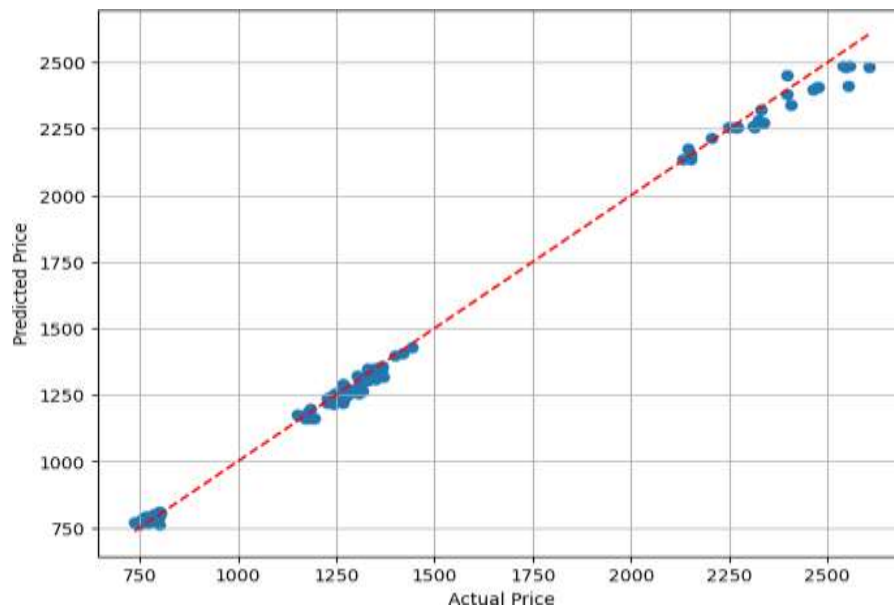


Fig 6. Actual vs Predicted Scatter Plot – Mid Stock Price

Figure 7 illustrates the accuracy comparison of different prediction models used for NIFTY-50 stock price forecasting. The results show that the proposed COAQNN model achieved the highest prediction accuracy of 98.27%, outperforming all other machine learning models. Among the baseline models, SVR obtained an accuracy of 92.52%, followed by Gradient Boosting with 87.96%, while Random Forest and Decision Tree achieved 80.56% and 80.52%, respectively. The superior performance of COAQNN demonstrates its ability to effectively capture complex nonlinear patterns and market dynamics, resulting in more accurate stock

price predictions. These findings confirm the effectiveness and reliability of the proposed COAQNN approach for financial forecasting applications.

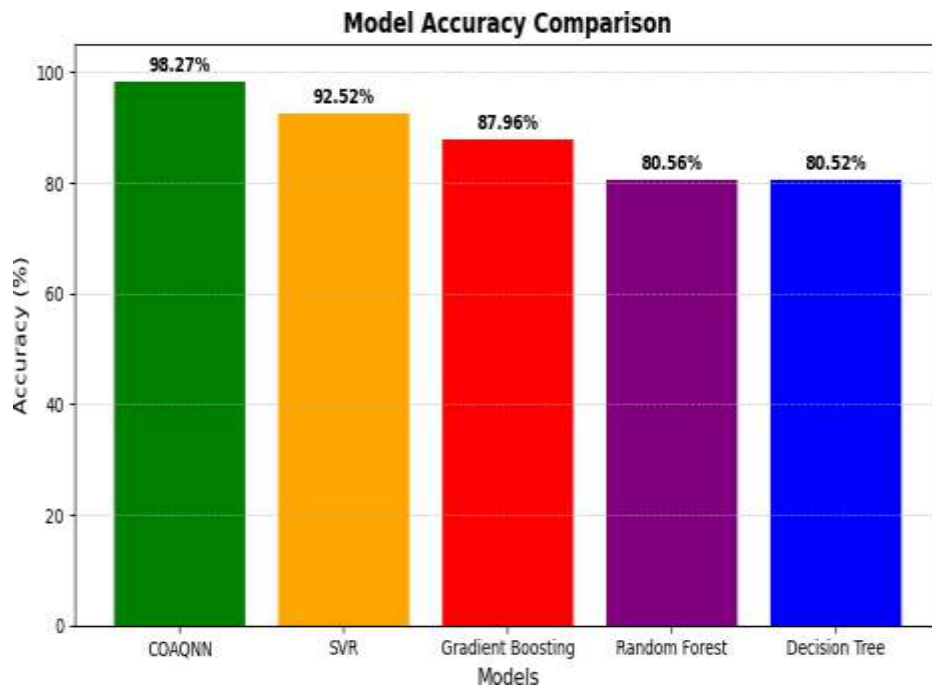


Fig 7. Model Accuracy Comparison

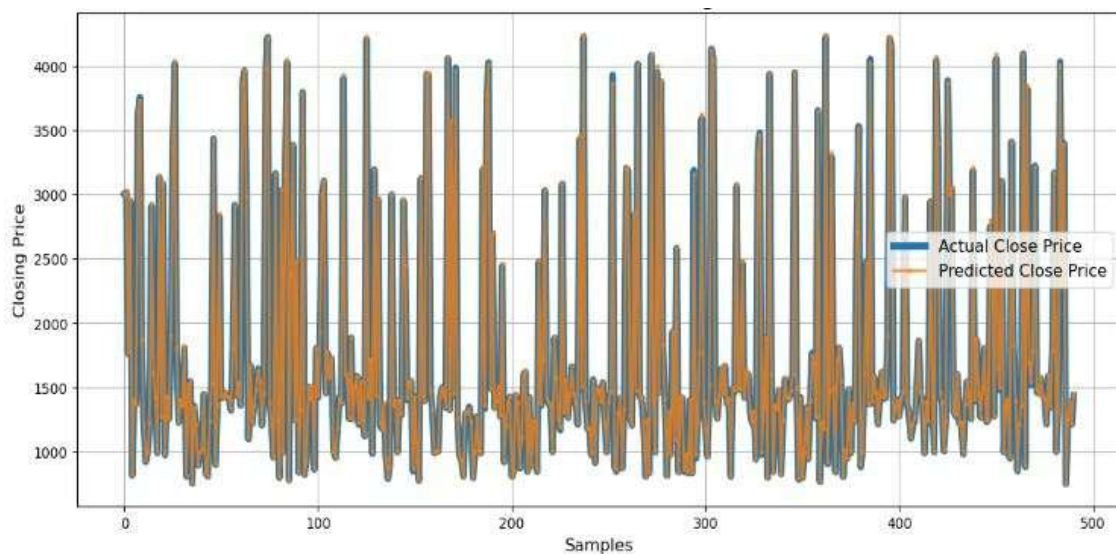


Fig 8. Actual vs Predicted – Closing Stock Price

Figures 9 and 10 illustrate the user interface of the proposed COAQNN-based Stock Price Prediction System for forecasting stock prices. Figure 9 demonstrates the prediction of the mid-day stock price, while Figure 10 presents the prediction of the closing stock price. In both cases, the user selects the desired stock, prediction date, and prediction type, after which the trained COAQNN model processes the input data and generates the corresponding forecasted stock price, providing valuable insights for investment and market analysis.



Stock Price Prediction System

COAQNN Based Stock Forecasting

Select Stock

RELIANCE.NS

Select Date

2026/05/22

Select Prediction Type

Mid Price

Predict Price

Predicted Mid Price: ₹ 1345.8599853515625

Fig 9. Midday Stock Price Prediction



Stock Price Prediction System

COAQNN Based Stock Forecasting

Select Stock

RELIANCE.NS

Select Date

2026/05/22

Select Prediction Type

Closing Price

Predict Price

Predicted Closing Price: ₹ 1358.9599609375

Fig 10. Closing Stock Price Prediction

V. Conclusion and future work

Predicting stock market prices accurately is a challenging task due to the highly volatile and non-linear nature of financial markets. To address this problem, the proposed research utilized a COAQNN-based deep learning framework for real-time NIFTY-50 stock price prediction. The developed system effectively predicts both mid stock prices and closing stock prices using historical market data, real-time financial information, and technical indicators such as RSI, EMA, MACD, volatility, moving averages, and lag features.

The proposed COAQNN model demonstrated high forecasting capability and achieved better performance compared to traditional machine learning models such as Random Forest Regression and Linear Regression. The system successfully minimized prediction errors and achieved an overall prediction accuracy of 98.27%. In addition, a Streamlit-based web application was developed to provide a user-friendly platform for real-time stock prediction and market analysis. The developed system is efficient, reliable, scalable, and capable of supporting investors and financial analysts in making informed investment decisions.

In future work, the proposed COAQNN-based stock price prediction system can be further enhanced by integrating advanced Natural Language Processing (NLP)-based news sentiment analysis, social media trend analysis, and transformer-based deep learning models such as GRU and Transformer networks to further improve forecasting accuracy and market trend analysis. The system can also be extended from NIFTY-50 stocks to global financial markets and integrated with automated trading systems, portfolio management modules, and cloud-based financial analytics platforms. Additionally, mobile application deployment, AI-based investment recommendation systems, and hybrid optimization techniques can improve scalability, real-time decision support, and practical applicability in intelligent financial forecasting environments.

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